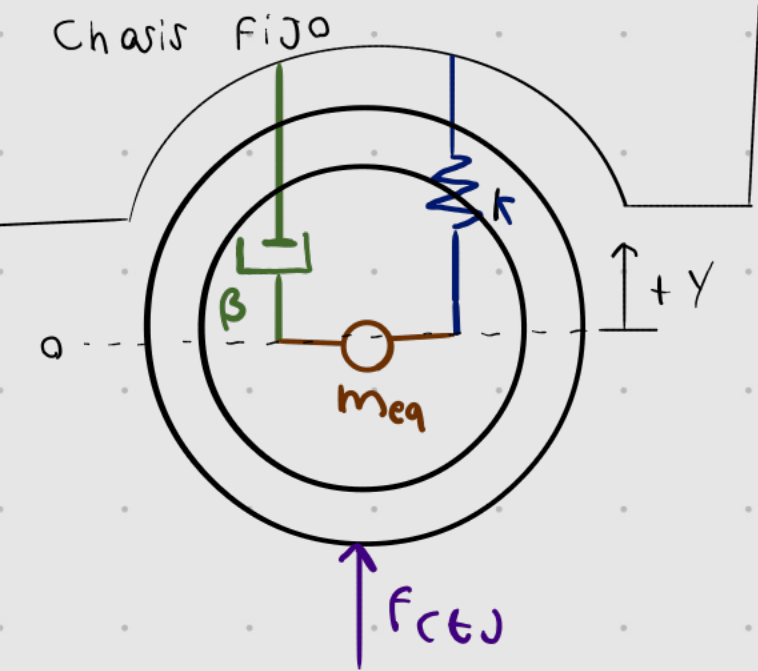




- Coordenadas generalizadas

solamente  $s_1$

$$q = y \rightarrow$$

desplaza  
verticalmente

- Identificando los términos del modelo

• Velocidad cinética  $T$

$$T = \frac{1}{2} m v^2 \rightarrow v = \dot{y} \rightarrow T = \frac{1}{2} m \dot{y}^2$$

• Energía potencial del resorte

$$V = \frac{1}{2} k y^2$$

• Función de disipación de Rayleigh

$$D = \frac{1}{2} \beta v^2 \rightarrow v = \dot{y} \rightarrow \frac{1}{2} \beta \dot{y}^2$$

• Lagrangiano

$$L = \frac{1}{2} m \dot{y}^2 - \frac{1}{2} k y^2 \quad \left. \begin{array}{l} \text{suma de} \\ \text{energías} \\ \text{cinéticas} \end{array} \right\} - \left. \begin{array}{l} \text{suma de} \\ \text{energías} \\ \text{potenciales} \end{array} \right.$$

## - Ecuación de Lagrange

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{y}} \right) - \frac{\partial L}{\partial y} + \frac{\partial D}{\partial \dot{y}} = Q$$

$$\frac{d}{dt} (m \dot{y}) - (-ky) + \beta \dot{y}$$

$$m \ddot{y} + \beta \dot{y} + k y = f_{ct}$$

Normalizando

$$\ddot{y} + \frac{\beta}{m} \dot{y} + \frac{k}{m} y = \frac{1}{m} f_{ct}$$

$\omega_n \rightarrow$  Frecuencia natural angular

$\zeta \rightarrow$  Razón de amortiguamiento



Comparamos con  $\rightarrow \ddot{y} + 2\zeta\omega_n \dot{y} + \omega_n^2 y = \text{entrada}$

De ahí que

$$\frac{\beta}{m} = 2 \zeta \omega_n$$

$$\frac{k}{m} = \omega_n^2 \rightarrow \boxed{\omega_n = \sqrt{\frac{k}{m}}}$$

$$\zeta = \frac{\beta}{2 m \omega_n}$$

$$\zeta = \frac{\beta}{2 \sqrt{m k}}$$

$$\zeta = \frac{\beta}{2 m \sqrt{k/m}}$$

$$\zeta = \frac{\beta}{2 m \frac{\sqrt{k}}{\sqrt{m}}}$$

$$\zeta = \frac{\beta}{2 (\sqrt{m})^2 \frac{\sqrt{k}}{\sqrt{m}}}$$

## Encontrando la función de Transferencia

$$y_{(0)} = 0 \quad \dot{y}_{(0)} = 0$$

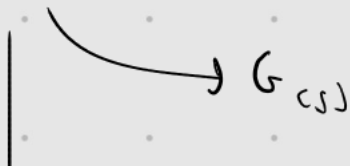
$$\mathcal{L}\{m\ddot{y}\} + \mathcal{L}\{\beta\dot{y}\} + \mathcal{L}\{ky\} = \mathcal{L}\{f_{(t)}\}$$

$$m(s^2 y_{(s)} - s y_{(0)} - \dot{y}_{(0)}) + \beta(s y_{(s)} - y_{(0)}) + k y_{(s)} = F_{(s)}$$

$$ms^2 y_{(s)} + \beta s y_{(s)} + k y_{(s)} = F_{(s)}$$

$$y_{(s)} (ms^2 + \beta s + k)$$

$$y_{(s)} = \frac{1}{ms^2 + \beta s + k} F_{(s)}$$


$$G_{(s)}$$

↓ Normalizando

$$G(s) = \frac{1/m}{s^2 + \frac{b}{m}s + \frac{k}{m}}$$

$$G(s) = \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad \left. \begin{array}{l} \text{Note que} \\ \frac{k}{m} = \omega_n^2 \end{array} \right\}$$

$$G(s) = \frac{(1/k)\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad \leftarrow \quad \frac{1}{m} = \frac{1}{k}\omega_n^2$$